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A PROJECT  
ON  
**Energy Efficiency in Multipath Routing Protocol on  
Wireless Sensor Network with Static Sink**

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# Contents

|                                                  |            |
|--------------------------------------------------|------------|
| <b>List of Figures</b>                           | <b>iii</b> |
| <b>List of Tables</b>                            | <b>iv</b>  |
| <b>Acronyms</b>                                  | <b>v</b>   |
| <b>1 Introduction</b>                            | <b>2</b>   |
| 1.1 Routing in Wireless Sensor Network . . . . . | 5          |
| 1.2 Applications of WSN . . . . .                | 6          |
| 1.3 Motivation . . . . .                         | 9          |
| 1.4 Objective . . . . .                          | 10         |
| <b>2 Literature Review</b>                       | <b>11</b>  |
| <b>3 The Proposed Scheme</b>                     | <b>17</b>  |
| 3.1 System model . . . . .                       | 18         |
| 3.1.1 Assumptions . . . . .                      | 18         |
| 3.1.2 Network Model . . . . .                    | 19         |
| 3.2 The Proposed Protocol (HSMRP) . . . . .      | 20         |
| 3.2.1 Neighbour Discovery . . . . .              | 21         |
| 3.2.2 Hop Number Assignment . . . . .            | 22         |
| 3.2.3 Multi-path construction . . . . .          | 24         |

|          |                                                 |           |
|----------|-------------------------------------------------|-----------|
| 3.2.4    | Data Transmission . . . . .                     | 28        |
| 3.3      | Flowchart . . . . .                             | 29        |
| <b>4</b> | <b>Simulation and Results</b>                   | <b>32</b> |
| 4.1      | Simulation Parameter . . . . .                  | 32        |
| 4.2      | Performance Parameters . . . . .                | 33        |
| 4.2.1    | Average Residual Energy . . . . .               | 33        |
| 4.2.2    | Number of live nodes . . . . .                  | 35        |
| 4.2.3    | Packet Delivery Ratio . . . . .                 | 37        |
| 4.3      | Advantages and Disadvantages of HSMRP . . . . . | 39        |
| <b>5</b> | <b>Conclusion</b>                               | <b>40</b> |
| <b>6</b> | <b>Future Scope</b>                             | <b>42</b> |
|          | <b>Bibliography</b>                             | <b>43</b> |

# List of Figures

|     |                                                   |    |
|-----|---------------------------------------------------|----|
| 3.1 | Random distribution of the sensor nodes . . . . . | 19 |
| 3.2 | Hop Number Assignment . . . . .                   | 23 |
| 3.3 | Multipath construction steps . . . . .            | 25 |
| 3.4 | Chart 1 . . . . .                                 | 29 |
| 3.5 | Chart 2 . . . . .                                 | 30 |
| 3.6 | Chart 3 . . . . .                                 | 31 |
| 4.1 | Average Residual Energy . . . . .                 | 33 |
| 4.2 | Number of live nodes . . . . .                    | 35 |
| 4.3 | Packet Delivery Ratio . . . . .                   | 37 |

# List of Tables

|     |                                                                   |    |
|-----|-------------------------------------------------------------------|----|
| 4.1 | Simulation Parameters Table . . . . .                             | 32 |
| 4.2 | Comparison Table on residual energy between MRP and HSMRP . . . . | 34 |
| 4.3 | Comparison table on number of live nodes between MRP and HSMRP .  | 36 |
| 4.4 | Comparison Table on packet delivery ratio between MRP and HSMRP . | 38 |

# Acronyms

**HSMRP** Hop count based ShortestRouting Protocol. 17

**MRP** Multipath Routing Protocol. 32

**WSN** Wireless Sensor network. 8

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# Abstract

In wireless sensor network(WSN), the sensor nodes are interconnected by wireless technique and communication among these nodes are done by routing protocols. Routing protocol is a standard that allows nodes to decide which route to take to route packets between the computing devices and is responsible for the end delivery of the data. The sensor nodes having limited battery life and transmission capability makes it hard to design an efficient routing protocol. The sink/base station of a WSN can either be static and mobile. In this paper, a hop count based shortest path multipath routing protocol(HSMRP) is proposed, which reduces the overhead of the network and increases the throughput of the network. The design protocol takes into consideration of flooding and unnecessary communication of nodes thereby increasing the residual energy, further improving the lifetime and reliability through discovering multipath from source to the destination. HSMRP is a sink initiated route discovery process, where source node location is known. In HSMRP, hop count is used and the shortest path amongst the multipath is chosen as primary path and the second next to primary path is chosen as alternate paths. Using these path for the transmission of data makes the system more efficient. Later, we analyze how the proposed protocol overcomes the drawback of the existing protocols.



# Chapter 1

## Introduction

A WSN is a network of large number of sensor nodes collectively deployed over a geographical area for monitoring physical phenomenon and environmental condition. The nodes work together and communicate by gathering information/data from a monitored field and then forward the data through multiple nodes up to the base station.

A wireless sensor network consists of two components that are Base station(sink) and Sensor node.

A Sensor node contains **micro sensors, microcontroller, radio transceiver, power source and memory** where, the microcontroller samples the micro sensor ; send the data, either with or without processing, through radio links to the locations where the information is needed.

- **Microcontroller:**

This is a chip which is very small in size, capable of doing powerful tasks including controlling the function of other devices connecting to it.

- **Transceiver:**

This is transmitter-receiver that is used for common purposes to send and receive data and commands.

- **External Memory:**

Sensor Nodes usually use flash memory due to small size and reasonable storage capacity.

- **Power Source:**

Power is stored in the form of batteries.

The sensor nodes of WSN has limited processing and transmitting capabilities; limited power, resource and storage. Generally, these sensor nodes are deployed in an isolated and hostile environment for monitoring battlefield, nuclear reactor etc. Therefore, replacing/recharging the battery of the nodes will be a difficult task. The sensor node not only sense environment but also gather data and forwards it to the base station. The base station also known as sink is also a kind of sensor node having rich resources, unlimited power, storage, better processors etc. It may be static or mobile depending on the scenarios. It communicates with the sensor nodes to collect the data and sends to the user.

A failure of some nodes in the network will cause network partition and part of the WSN will be cut off from the rest of the network. Network lifetime is therefore, dependent on the lifetime of individual sensor nodes. This raises the issue of energy efficiency design in the network. Thus, the network requires an energy efficient **routing protocol** to enhance the network lifetime.

In Wireless Sensor Network designing an efficient routing algorithms for different purpose, applications, fulfilling the different performance demands is considered as an important issues. To handle such issues many routing algorithms have been proposed in order to enhance the performance demands of various applications through the network layer of the wireless sensor networks protocol stack. But most of the algorithms are based on single-path routing. Basically in single path routing approach the source node selects a single path which will satisfy the performance demands of the application for transmission of data towards the sink.

Even though the single path between the source and base station/sink can be developed with minimum computation complexity and resource utilization, the other factors such as the limited capacity of single path reduces the available throughput. Secondly, the single path routing is not flexible to link failures considering the unreliable wireless links, thus degrading the performance of the network .

And then finding a new primary path after the failure of the existing primary path to continue the data transmission will only cause an extra overhead and increase delay in data delivery. Due to these issues/factors we cannot consider single path routing as an effective technique to meet the performance demands of various applications.

Multi-path routing strategy protocol is introduced in order to overcome the performance issues and the limitations of the single path routing strategy. The Multi-path routing is also known as alternate path routing strategy. In multi-path routing strategy, multiple path will be established between the source and the sink/base station aside the primary path and the multiple path will be act as an alternate path for the transmission of data. The usages of these alternate path are based on the individual routing strategy. Like in some routing algorithms, the best path is used to send the data while the alternate paths is used when there occurs a failure to the primary path as a backup and some use all the paths concurrently to send data and so on.

In the past few years multi-path routing approach is extensively used for different network management purposes, such as providing a fault tolerant routing, improving transmission reliability, congestion control and Quality of Service(QoS) support in the wired and wireless networks, but the unique features of the wireless sensor networks and the characteristics of the short range radio communications introduce a new challenges that should be addressed in designing the multi-path routing protocols.

## 1.1 Routing in Wireless Sensor Network

In the wireless sensor network, routing technique plays an important role[1]. As WSN has limited energy, storage and processing capability, it is a very challenging task to develop a routing protocol. In most of the case, multiple sources are required to send their data to a particular base station/sink[2]. The nodes nearer to the sink tends to die early as it used for every transmission. This causes partitioning of the network consequently, reducing the lifetime of the network. The main constraint of the sensor node is energy. The sensors are battery-powered computing devices. It is hard to replace the batteries in many applications. Therefore, WSN requires an energy efficient protocol that will consume less power during the process. Due to dense deployment, the sensor nodes generate redundant data, and the base station may receive multiple copies of the same data. Therefore, energy of the sensor nodes are consumed unnecessarily. WSN is highly dynamic and does not have fixed infrastructure .

There are two main reasons responsible for the dynamic infrastructure. The first reason is the energy; the sensor nodes are equipped with limited energy in the form of batteries. The sensor node might die if the protocol is unable to balance the load among the nodes leading to the dynamic network structure. The second reason is the mobility; in many cases after the sensor nodes are deployed, the nodes are static but sink can move within the network, making the network dynamic, and the protocol that works for static sink might not work for mobile sink[3].

Since the main task of the wireless sensor networks is the transmission of data from the source to the sink, and the data forwarding method is an important issue which should be considered in developing these networks. Taking into consideration of low power in wireless sensor networks, routing in WSN is much more challenging compared to traditional wireless networks such as ad-hoc networks.. Firstly, as the nodes are highly dense , the routing protocols should route data over long distances without taking into

consideration of the network structure and size, besides the above requirement failure of active sensor nodes during operation may occur due to the environmental factors or energy depletion of sensor nodes or hardware faults, but these cases should not interrupt the normal operations of the network. Since, the wireless sensor nodes are restricted in terms of power supply, processing capability and available bandwidth, routing and data forwarding should be performed with effective network resource utilization. Further, considering the performance demands of the wireless sensor networks are totally application dependent, routing algorithms should satisfy the demands of the application for which the network is being deployed.

## 1.2 Applications of WSN

Wireless Sensor network consist of sensor nodes deployed over a geographical area for monitoring physical phenomenon like Area monitoring, Air pollution monitoring, Forest fires detections, Landslide detection and some other commercial aspects like Greenhouse monitoring, structural monitoring, Agriculture.

- AREA MONITORING

WSN nodes are deployed over a region where some phenomenon is to be monitored. Where the sensor nodes detects event being monitored (heat,pressure), reports to one of the base station, to take up action. E.g., Military to detect enemy intrusion and civilian: geo-fencing of gas or oil pipelines.

- HEALTH CARE MONITORING

There are several types of sensor networks for medical applications: implanted, wearable, and environment-embedded. Implantable medical devices are those that are inserted inside the human body. Wearable devices are used on the body surface of a human or just at close proximity of the user. Environment-embedded

systems are body position measurement, location of persons and physical state, overall monitoring of ill patients in hospitals and at home.

- AIR POLLUTION MONITORING

WSNs have been deployed at various cities to monitor the concentration of dangerous gases for citizens.

- FOREST FIRE DETECTION

Sensors nodes are deployed in forests to detect the location of the fire and send the information to the end user to control fires where the nodes equipped with sensors to sense temperature, humidity and gases.

- LANDSLIDE DETECTION

Make use of the WSNs to detect slight soil movement and changes in various parameters that may occur before or during a landslide.

- GREENHOUSE MONITORING

WSNs are used to control temperature, humidity levels in commercial greenhouses. When any fluctuation in temperature and humidity levels. The sensor nodes will send emails/text messages, or trigger systems, open vents, turn on fan etc.

- AGRICULTURE

Some uses of WSNs in Agriculture are frees from wiring in difficult environment, Gravity feed water systems to water level, pumps can be controlled and Reduces wastage of water b the used of sensor node.

- STRUCTURAL MONITORING

In WSNs, Sensor nodes monitor the movement within buildings and infrastructure such as bridges, flyovers, embankments, tunnels etc. Which reduces costly sites visits, as well as advantage of daily data collection more efficiently and effectively.

## 1.3 Motivation

A wireless sensor network is a network that are form by randomly distributed sensor nodes interconnected with one another to sense and monitor physical or environmental conditions. Sensor nodes are run by battery and the batteries cannot be replaced. The nodes die when the battery dies affecting the functionality of the network. Thus, a routing technique is very much essential to enhance the life span and manages the battery efficiently. This characteristic motivates to design energy-efficient routing techniques. Wireless sensor network is a multi-hop network where data are transmitted through the intermediate sensor nodes. The links between sensor nodes are highly prone to failure. The frequency of link failure directly affects the data delivery ratio and decreases the reliability of the network. This issue motivates to design reliable routing techniques. Wireless sensor network is a multi-hop network where data are transmitted through the intermediate sensor nodes. The links between sensor nodes are highly prone to failure. The frequency of link failure directly affects the data delivery ratio and decreases the reliability of the network. This issue motivates to design reliable routing techniques. The power issue in the wireless sensor network is one of the biggest challenges, because the sensor has a limited source of power which is also hard to replace or recharge .Apart from Delay and Throughput, power consumption is one of the main performance parameter of a WSN routing protocol .To improve the performance parameter of such networks more efficient protocols are needed.



## 1.4 Objective

The main objective of the multipath routing is to provide reliability to the network through available paths between a sensor and the sink. And also to analyse the multipath routing protocol where forwarding packets is an important research in wireless sensor networks which have a restricted communication range and high density of sensor node. So, multipath routing protocol is widely used to improve network performance by discovery multiple path and through efficient utilization of the available network resource. The main aim of this survey is to present the concept of multi-path routing, as well as the basic motivations for utilizing this technique which would help to conserve the energy available in each sensor node by increasing network lifetime which is important in wireless sensor networks where the multi-path routing algorithm would help to increase the network lifetime thereby the efficiency of the nodes in it and also provides an effective way to prolong the lifetime of the network. This paper also focus on the objective behind the development of each protocol category. Finally, a detailed study has been made on the design and operation of the different protocols and on the proposed new protocol and brief analysis of the simulation results against the already known metrics with network and energy lifetime being major among them.

# Chapter 2

## Literature Review

### **Multi-path Routing protocol with static sink:**

Multi-path-based Routing Multi-path routing is an alternative routing technique, which selects multiple paths to deliver data from source to destination. It allows multiple paths between the source and the sink. Due to the use of redundant paths, multi-path routing can largely address the reliability and load balancing issues. Here, all the sensor nodes have the same resources and capability. Many multi-path routing protocols have been proposed for WSNs.

- **Low Interference Energy-efficient Multi-path Routing (LIEMRO)**

Low Interference Energy-efficient Multi-path Routing (LIEMRO) [4] is proposed by Radi et al. In LIEMRO node-disjoint paths is used which improves the latency, lifetime and packet delivery ratio. It includes a load balancing algorithm to distribute the source node traffic over multiple paths based on the relative quality of each path. In LIEMRO, the Expected Transmission Count (ETX) metric is used to calculate the cost of the links. In this method, the sink sets its cost to zero and broadcast a control packet to its neighbors. Each neighbor then calculates its link cost with respect to the sink. Further, they broadcast the information in the network until the source node receives the information. As soon as an event

is triggered, the route discovery phase is initialized in the network. Then for the route establishment to start, the source node sends the route request to the sink. The path having with lesser residual energy transmits the data with a lower rate to save the energy. Based on the quality of the paths LIEMRO conserves the traffic rate dynamically. However, it does not consider the service rate and the buffer capacity of the active nodes to adjust and predict the traffic rate of the active paths.

- **Maximally Radio-disjoint multi-path routing (MR2)**

**Maximally Radio-disjoint multi-path routing (MR2)** [5] is proposed by **M.Maimour**, which deals with the problem of both inter-session and intra-session interference. Providing the necessary bandwidth to multimedia data through non-interfering paths is the main objective of MR2. Minimum interfering paths is constructed using the adaptive incremental approach/method to satisfy the bandwidth requirement of the multimedia. With the incremental approach it constructs paths from the source to the sink and only one path is constructed at a time and the other additional paths are constructed whenever cases of network congestion or bandwidth shortage arises. Interference awareness can be attained by keeping parts of the sensor nodes in sleep mode/state. During the sleep state, those sensors will not take part in any routing process. Here, like other query based routing protocol, the network uses flooding to send the route request message from the sink node in order to start the route discovery process. On receiving the route request by the immediate neighbour nodes of the base station/sink node, the id of the respective nodes is added to the received request as their path id and rebroadcast the message. Whenever a node receives a route request message it checks whether the requested id is already in the routing table of the reported path, if any path does not exist between the source node and the sink node for new route, it should add the reported path ID in its routing table, otherwise if the requested

id is already a part of the routing table it is the updated in the routing table if it provides a better metric than the previous one. This process continues until the source which contains the requested data receives the request message. After receiving the request message, the source transmits data through the best metric path. To tackle the interference problem those nodes of active path notify its intermediate nodes to be in sleep mode/state, in order to prevent them from taking part in the route discovery process. Therefore, during data transmission process the nodes which receive the data packet should send a be passive message to all its neighbors except from the source from where the message has arrived and the immediate next-hop node along the active path. This mechanism provide sufficient bandwidth for data transmission as the passive nodes are not able to answer to any request message. Even though this protocol get rid of the negative effects of the interference by putting a subset of nodes in passive state. The main drawback of this protocol is that it is only suitable for query driven application, secondly, the flooding technique used in the route discovery process imposes a high control overhead.

- **Ye Ming Lu et al** have proposed a distributed, scalable and localized routing algorithm . In this protocol, between the sink and the source nodes multiple node-disjoint paths are discovered. It also distributes the traffic over the multiple paths using a load balancing algorithm. A node is selected from the area where an event is detected as a source node. The source node then starts the route discovery process. The sink sends multiple route request messages to its neighboring nodes with distinct path id to build node-disjoint paths. After receiving the first route request message from the source node, a timer is set by the sink. Any path discovered after the timer stops are discarded. The sink also optimally assigns the data rate for each path.

- **Interference-Minimized Multi-path Routing Protocol (I2MR)**

Jenn Yue Teo; Yajun Ha; Chen Khong Tham [6] proposed the **Interference-Minimized Multi-path Routing Protocol (I2MR)** which aims to support high data rate streaming in wireless sensor networks taking into account of the recent advances in designing the high bandwidth backbone network. I2MR assuming a special network structure and tries to construct zone disjoint paths and distributes the traffic over the discovered paths with the availability of a particular hardware component. All the deployed gateway nodes are assumed as the final destinations and it is assumed that these nodes are directly connected to the command center using non-interfering and high capacity links. Two paths are constructed by the source node which will be used to transfer data and keeps only one backup path to the central command center.

There three main steps in the route discovery process.

1. One gateway node as its primary gateway node is selected for each source and the shortest path towards this gateway node is constructed
2. Marking steps of all the one hop and two hop neighbors of the intermediate nodes along the first path are marked as the interference zone of the primary path.
3. In the last step, the secondary and backup gateway nodes is selected by the primary gateway which determines the preferred quadrants with which the secondary path is selected.

For path construction paths between each source node and the command center, the quadrants are decided based on the location of the source node, further the preferred gateway nodes and should have less distances to the source node than the other candidate gateway nodes be located beyond the interference range of the primary gateway. As for the secondary path backup nodes are determined, the source starts the construction of the secondary and the backup paths using the nodes that are not marked in the interference range of the primary path.

After the construction of the path is completed the source node uses the primary and secondary paths with the highest possible data rate and keeps the third path to achieve prompt recovery upon path failure. During data transmission if any defect, long congestion occurs in the intermediate node then it should notify the source to reduce the assigned data rate. The I2MR have performance better in comparison to other multi path routing protocol, but the achieved improvements require special network structure and particular hardware support which may not be feasible for many applications. And also the zone-marking algorithm adds high complexity to the system, due to which interference-minimized paths cannot be constructed effectively by this method. Furthermore, the source node creates three separate shortest paths to reduce wireless interference among the successive nodes along a path towards three different gateways. Nonetheless due to irregular time changing properties of low power wireless links, transmission of data over a long hops results in increased packet loss rate

- **Energy Efficient and Collision Aware Multi-path Routing Protocol**

Energy Efficient and Collision Aware Multi-path Routing Protocol is a multi-path routing protocol which uses information of the location of the nodes to establish two collision free paths on both sides of the direct line between the source-sink pair. Interference effects are reduced by keeping the distance between the two paths more than the interference range of the sensor nodes. Here, two distinct set of nodes on both sides of the direct line between the source-sink pair are discovered by the source in the first stage. The source node after finding the neighbor set broadcasts a Route-request packet towards these nodes to discover two node disjoint paths. During this step the intermediate nodes also follow the same technique as the source node to select there next hop. Upon reception of a route request message the intermediate node set a back-off timer to restrict the overhead of the route discovery flooding. Before broadcasting the received route request message the

intermediate nodes set a back-off timer based on their distance to the sink and residual battery level. The neighbor nodes having high battery level and shorter distance to sink are set with shorter back-off time , therefore, at each stage to broadcast the received Route-request packet towards the sink node only a single node achieve to do so. The sink node on receiving the route request sends a route reply in the reverse path towards the source. When the source receives the route reply it starts the data transmission through the established path. Even though two shortest non-interfering paths from the source to base station/sink is discovered in EECA but still it needs the sensors to be assisted by The GPS and depends on the information given by the underlying localization update method. These requirements increase the cost of the deploying the network and also increase the communication overhead, specially in dense and large wireless sensor networks.

Secondly, the low-power wireless links exhibit signal variation over time, calculating the interference range of the sensors based on distance may not provide accurate result of interference estimation. Thirdly, transmitting data over minimum hop paths can reduce delay and resource utilization but using these paths can result in packet loss and an overhead of packet re-transmission.

# Chapter 3

## The Proposed Scheme

In this paper, we proposed a protocol based on the multi-path routing scheme where multiple routes exist between each source and the sink and if one path fails among those routes, data is sent through another different available path. Thus increasing the reliability of the network system. A system of multi-path routing is reliable if it achieves to maintain the routing path during data transmission and the data must reach the sink through a reliable path. Multi-path construction is possible due to dense deployment of sensor nodes

In this paper, a hop count based shortest-path multi-path routing protocol (HSMRP) is proposed, which aims to conserve energy [7] and find reliable multi paths. It also reduces the control overhead of route discovery and increases the throughput of the network. It is based on the existing protocol MRP. HSMRP is the sink initiated route discovery process with the known location information of the source node.

In this protocol, we introduce hop count for finding the shortest path among the multiple paths discovered from the source to the sink in which the MRP fails to do so. In MRP, primary and alternate paths are not discovered based on the shortest path from the source to the sink. Whereas in HSMRP the shortest path from the source to the sink



is chosen as the primary path using smallest hop count. Likewise, the second shortest path among the multiple path discovered is chosen as the alternate path. Unlike other protocol having many alternate paths, HSMRP have the shortest alternate path which leads to improvement in the lifetime, latency and reliability through discovering multiple shortest paths from the source node to the sink. In the following section, we will discuss system model and assumptions for the proposed protocol and workings of the proposed protocol

## 3.1 System model

### 3.1.1 Assumptions

The following assumptions are considered for the proposed protocol:

- Sensor nodes and sink (i.e base station) are deployed randomly in the environment and form a network.
- Sensor nodes and sink are static after deployment.
- The sink have unlimited memory, computation and battery power as compared to the sensor nodes.
- The nodes have limited transmission range.
- All the sensor nodes are homogeneous and have the same capabilities.
- Sensor nodes can calculate their residual energy.
- Communication range of each nodes are identical and predefined.

### 3.1.2 Network Model

Here, the network consists of  $n$  number of sensor nodes and a base station/sink node. The sensor nodes are deployed randomly into the environment. All the sensor nodes know their position information, and the sink node contains the *id* and *location info* of each sensor node. When the sink requires the data from the source node, it constructs the route between them. For path discovery, hop count is used [8]. The threshold energy is the minimum residual energy of a sensor node, beyond which it cannot perform any additional functions except sensing and relaying the data.

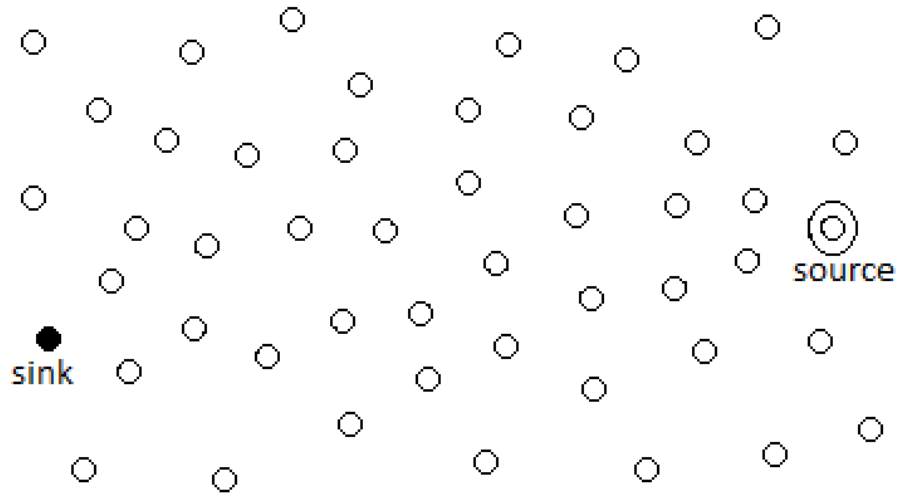


Figure 3.1: Random distribution of the sensor nodes

## 3.2 The Proposed Protocol (HSMRP)

With the above assumptions, the working principle of HSMRP is presented in this section. HSMRP avoids the flooding and takes the benefit of both load balancing and collision aware mechanism for energy conservation of the network. The protocol finds the shortest distance between the source and the sink using shortest hop count value. It mainly consists of four phases: neighbor discovery, hop count determination, multi-path construction, data transmission.

### 3.2.1 Neighbour Discovery

In this phase, finding of the neighbouring nodes and maintenance of neighbour's information takes place and is illustrated in the Algorithm 1. Here, an initiator node broadcast a control packet which contains the id of the nodes, residual energy and the location of the node. As the residual energy of all the nodes are the same, the initiator nodes are chosen randomly. The neighbour nodes which receives the packet will be appended to a table called NbrTable. The NbrTable contains the id of the sensor nodes, residual energy and the location information. The recipient node drop out the packet of the the sender node if the sender node's id is already available in the NbrTable. The control packet will be broadcast by the recipient nodes if it does not broadcast before. After the neighbor discovery phase each nodes will have a list of its neighbor nodes.

---

**Algorithm 1** Algorithm for Neighbour discovery:

---

**Data:-**  $sn[]$  = list of sensor nodes.

$i$  = sensor nodes in the list.

$n$  = number of sensor node.

    NbrTable = neighbour table of  $n$  nodes.

    energy: residual energy of node  $i$ .

**OUTPUT:-**

```
1: function neighbor_discovery(n):
2:   for i in range(0,n): do
3:     energy(sn[i])
4:     for j in range(0,n): do
5:       if sn[j].dead == False: then
6:         calculate distance d
7:         if d < radio_range and i ≠ j and n[i].dead == False: then
8:           sn[i].NbrTable.append(sn[j])
9:         end if
10:      end if
11:    end for
12:  end for
```

---

### 3.2.2 Hop Number Assignment

In this phase, hop number assignment takes place where each sensor nodes are assign with hop number. The hop numbers are then used to discover paths where transmission of data will take place. As it is sink initiate route discovery process , the hop count starts from base station upto the source node. Each sensor node have a radio range, energy-level, NbrTable and hop-no. The base station will transmit the broadcast and the sensor nodes which are within the radio range of the base station will have their hop-number unit as 1. Likewise, the nodes which are within the radio ranges of nodes which have 'i' as their hop-no becomes 'i+1'. The hop-no are assigned to each nodes in the similar manner and is illustrated in the Algorithm 2. These hop-no. will be used to determine the shortest path from the source to the sink. And whenever a node receives the broadcast message it checks whether the neighbour node's id is already in the routing neighbour table, if not then hop count assignment will occur but if the id is already in the table and the hop count from the new nodes is better than the previous existing hop count then the routing table is updated to the better count.

---

**Algorithm 2** Algorithm for Hop Number Assignment:

---

**INPUT:-** ss=list of sensor nodes

NbrTable= neighbour discovery table of n nodes.

**OUTPUT:-**

```
1: Function hop_assign(ss):
2:   for node in ss.NbrTable: do
3:     if node.hop == -1 or node.hop > ss.hop + 1: then
4:       node.hop = ss.hop + 1
5:       hop_assign(node)
6:     end if
7:   end for
8:   for i in range(0,n): do
9:     if sn[i].hop == 0: then
10:      sn[i].hop = 1000;
11:    end if
12:   end for
13: sn[0].hop = 0
```

---

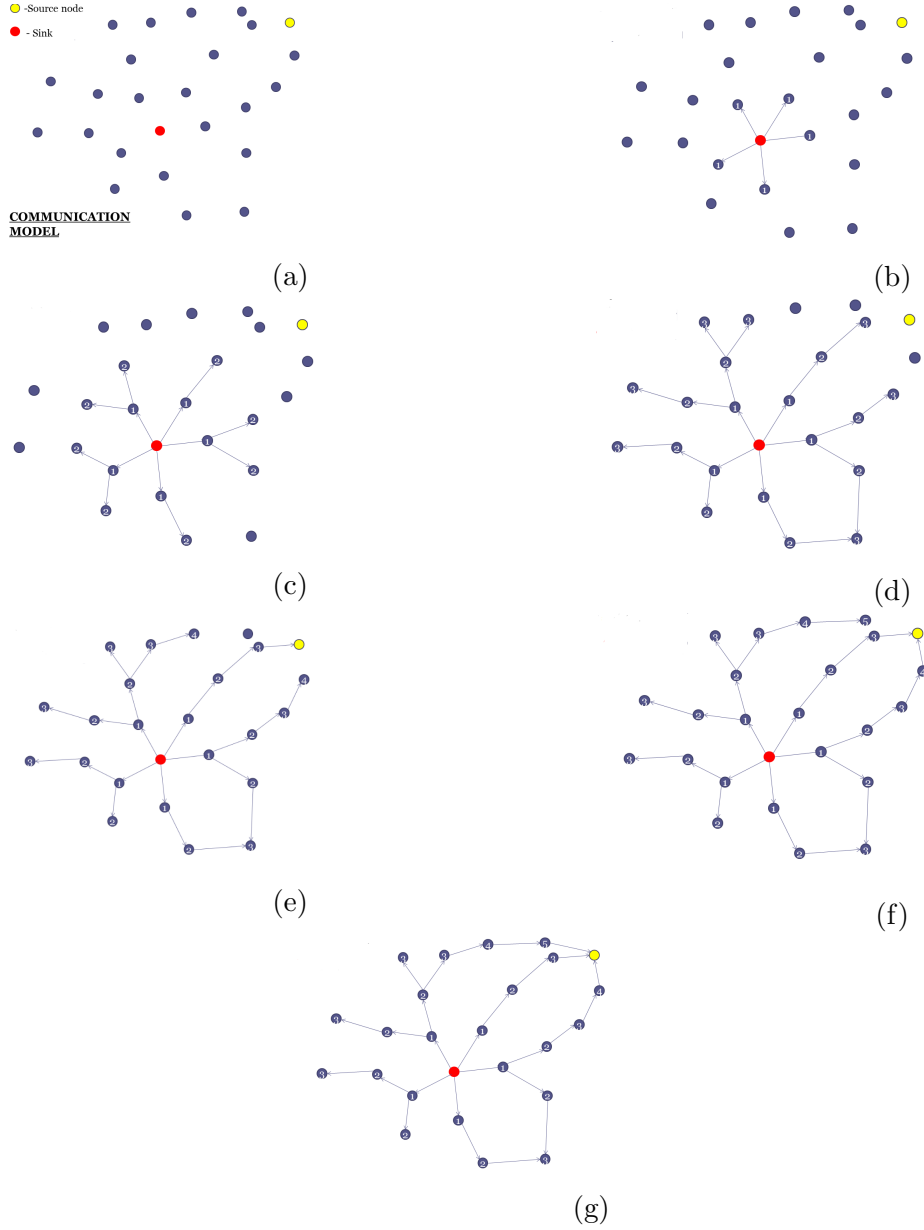


Figure 3.2: Hop Number Assignment

The above figure shows the illustrations of how hop count works. Here, the sink/base station sets its own hop count to 0 and the nodes which are within the radio range of the sink/base station receive a packet and are set to a hop count of 1. And then those nodes which are having hop count of 1 then send a packet to their neighbouring nodes which are within their radio range and then set the hop count to the previous hop count + 1. Similarly in this manner the remaining nodes are set to hop count in an

increasing manner until it reaches the source node.

### 3.2.3 Multi-path construction

After the neighbor discovery phase and hop number assignment phase, each node possesses their neighbor information and then construction of Multi-path phase will start. It is assumed that the location of the base station/sink is known by the source node in prior. The route discovery based on the location of the source will be initiated by the sink. And using the above hop number assignment algorithm, the sensor nodes are assigned with hop numbers. These hop number are then used to create paths where data from the source node will be transmitted to the sink. The path having the smallest total hop count from sink to the source node will have the shortest path while transmitting the data from source node to the sink. This definition is recursively used in the algorithm. The primary path is constructed with the shortest route from the sink to the source node(shortest hop count path) and the alternate path is constructed with the next shortest path. One primary path and multiple alternate paths are constructed between the sink and the source node. The paths are partially node-disjoint in the HSMRP. The process of multipath construction is presented in the Algorithm 3 and 4. The Algorithm 3 and 4 procedures `path_primary_create[]` and `path_alterate_create[]` are repeated untill data transmission is completed for each round.

- Find PrimaryPath[]: This function is called to find the primary path. The nodes of the route having the shortest hop count from the sink to the source is chosen as the primary route/path for the transmission of data. The primary path is known using hop-count where the id of the next nodes are found using Nbrtable and sent packet to inform the neighbors. Then it intimates about the status of the next node by unicasting a message. Similarly, the alternate route having the second shortest path will replace this primary path if any disruption occurs.

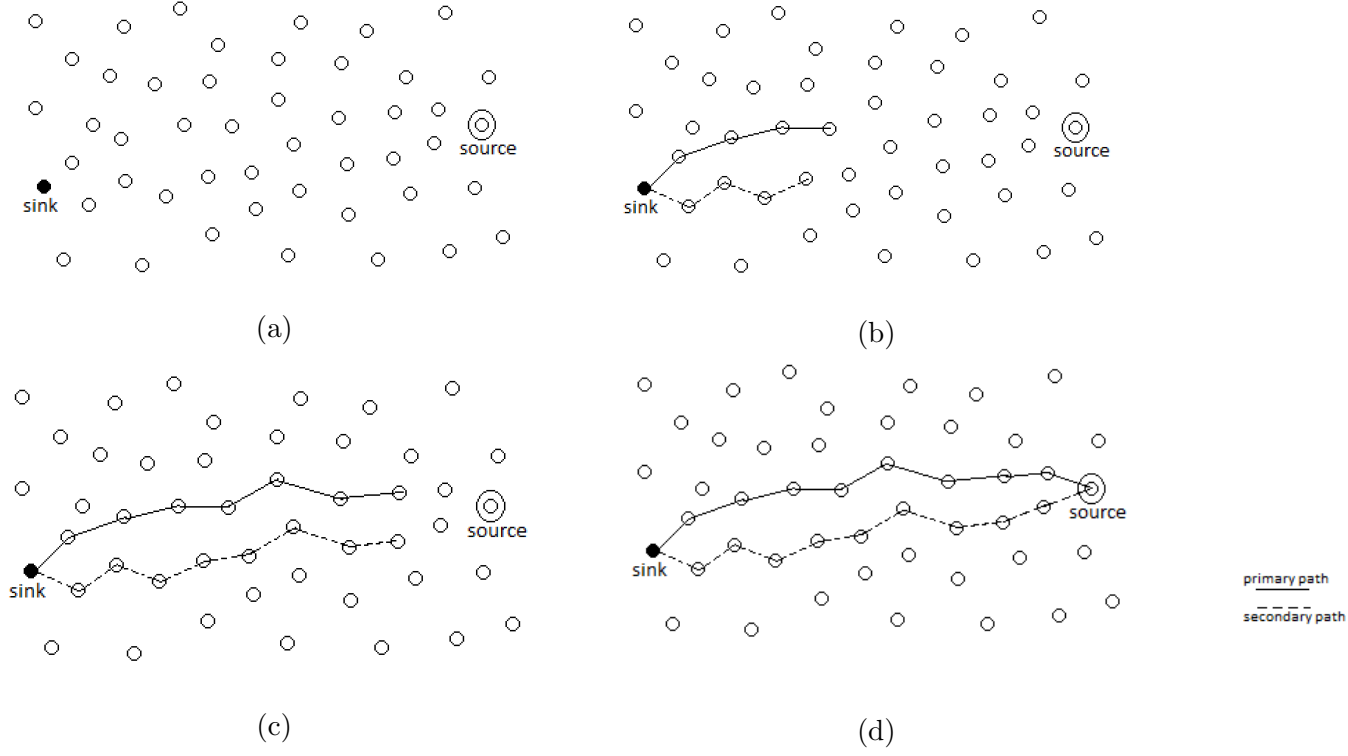


Figure 3.3: Multipath construction steps

- Find AlternatePath[]: This function is called to find an alternate path which is the second best route after primary path. The path having the second least hope count value is chosen as the alternate path. All the nodes except the primary nodes are switched to sleep mode. At a time, only one path will be active between the source node and the sink. This reduces the interference from any other path and avoids the collision. Both these factors contribute to reducing the energy consumption. If the primary path gets disrupted, the protocol selects the alternate path to transmit data. However, if all the paths are disrupted, then the routing process starts over again from the neighbor discovery phase.



---

**Algorithm 3** Algorithm for Primary Path Discover:

---

**INPUT:-** path\_primary[] =list of node that will be used in primary path creation.

NbrTable= neighbour discovery table of n nodes.

Hop= hop count.

**OUTPUT:-**

```
1: Function path_primary_create(x):
2:   path_primary.append(x)
3:   current = path_primary[len(path_primary)-1]
4:   while (current.id != 0:) do
5:     for nn in current.NbrTable: do
6:       if nn.hop == current.hop - 1: then
7:         path_primary.append(nn)
8:         energy(current)
9:         current = path_primary[len(path_primary) - 1]
10:        if current.id == 0: then
11:          Break
12:        end if
13:      end if
14:    end for
15:    current = path_primary[len(path_primary) - 1]
16:  end while
```

---

---

**Algorithm 4** Algoritthm for Alternate Path Construction Part 1

---

**INPUT:-** path\_alternate[] =list of node that will be used in alternate path creation.

NbrTable= neighbour discovery table of n nodes.

Hop= hop count.

**OUTPUT:-**

```
1: Function pat_alternate_create(x):
2:   if len(x.NbrTable) == 0: then
3:     path_alternate.append(x);
4:   else:
5:     alt = x.NbrTable[1] path_alternate.append(x);
6:     path_alternate.append(alt);
7:   end if
```

---

---

**Algorithm 4 (Continued) Part 2**

---

```
8: current = path_alternate[len(path_alternate) - 1]
9: while (current.id != 0): do
10:   for nn in current.NbrTable: do
11:     if nn.hop == current.hop - 1: then
12:       path_alternate.append(nn)
13:       current = path_alternate[len(path_alternate) - 1]
14:       energy(current)
15:       if current.id == 0: then
16:         break;
17:       end if
18:       current = path_alternate[len(path_alternate) - 1]
19:     end if
20:   end for
21: end while
```

---

In the algorithm all nodes except the primary nodes are put to sleep mode. At a time there is only one active path between the source node and the sink node. This is done to reduce interference from other paths and avoid collision. There arises some cases where there can exist

### 3.2.4 Data Transmission

After the route discovery by multipath construction[9] [10], transmission of data will take place where the data is transmitted between the source and the sink. Among the two path created i.e the primary and the alternate path, the primary path is chosen to transmit data[11]. When the primary path disrupt or fails the alternate path will be used instead for data transmission. So, in general transmission [12] of data from the source node takes place over the primary path and the alternate paths are used when the primary path is not available. And if path does not exist between the source and the base station/sink node, then the routing process starts once again. The remaining nodes that are not part for primary and alternate path creation will go to sleep mode in order to conserve energy.

### 3.3 Flowchart

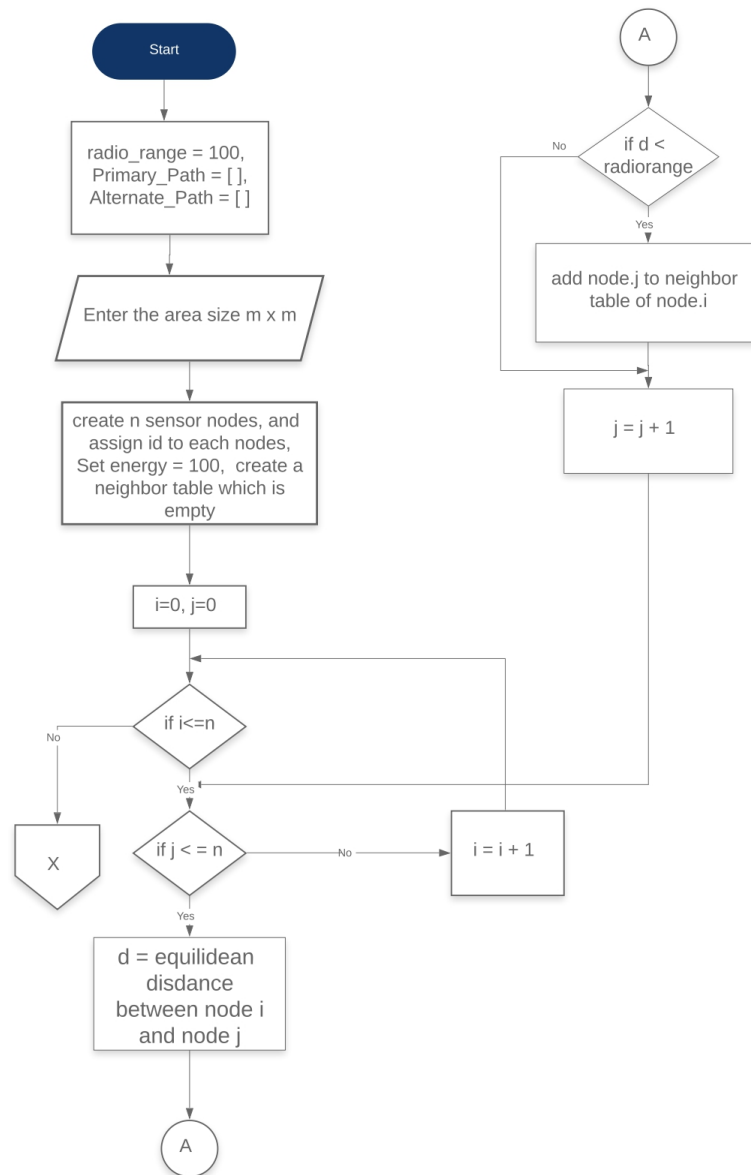


Figure 3.4: Chart 1

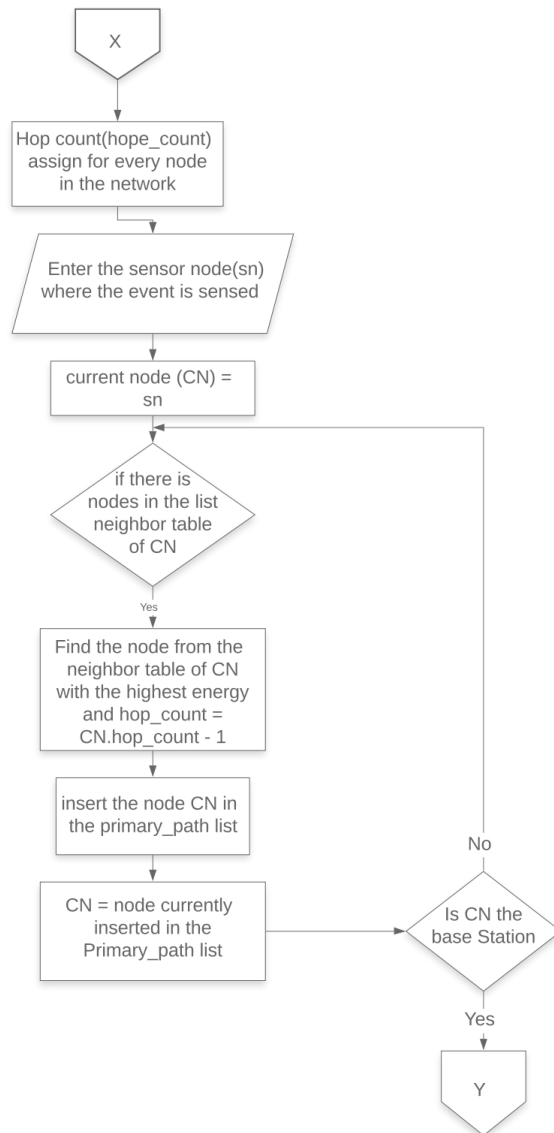


Figure 3.5: Chart 2

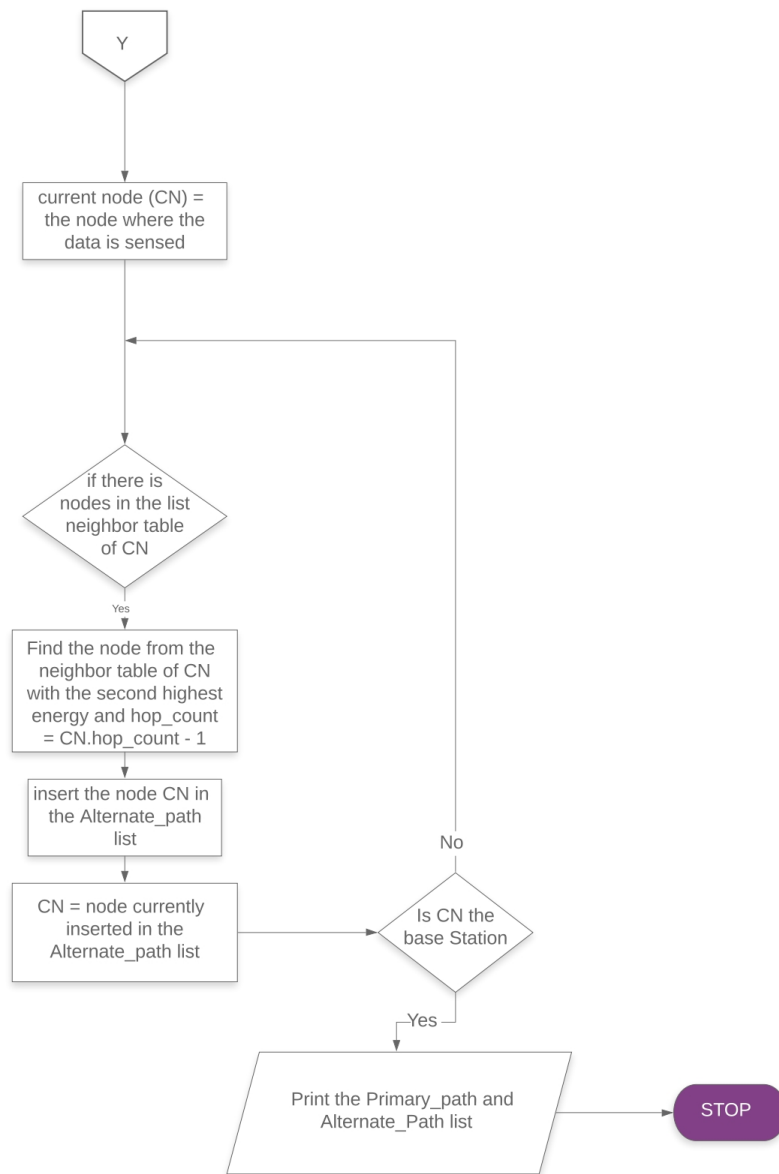


Figure 3.6: Chart 3

# Chapter 4

## Simulation and Results

### 4.1 Simulation Parameter

The proposed protocol HSMRP is implemented in Python. It is a simulator for WSN based upon the OMNeT++ platform [99]. The Multipath Routing Protocol(MRP) [13] is also implemented and considered the following simulation parameters as mentioned in the table below. The simulation parameters are listed in the Table 1

| Parameter Name         | Value                        |
|------------------------|------------------------------|
| Simulator              | Python                       |
| Simulation Area        | $Y \times Y \text{ meter}^2$ |
| Channel                | Wireless                     |
| Number of nodes        | 100                          |
| Radio range            | 100                          |
| Intial Energy          | 1                            |
| Intial Hop value       | -1                           |
| Intial Residual Energy | 100                          |
| Size of packet         | 32byte                       |

Table 4.1: Simulation Parameters Table

## 4.2 Performance Parameters

### 4.2.1 Average Residual Energy

It is the average of amount of energy which remains in each nodes after transmitting data from the source to the sink[14] As the sensor nodes in the WSN have limited resource , energy consumption by the nodes became an important aspect for the network lifetime. Here, we will look at the amount of energy left on average of the network after some round of data transmission. The result is shown in the figure below. From the figure we can see that the existing protocol consume more energy thus giving less residual compare to that of the proposed protocol. The proposed protocol has more residual energy compare to both the existing protocol.

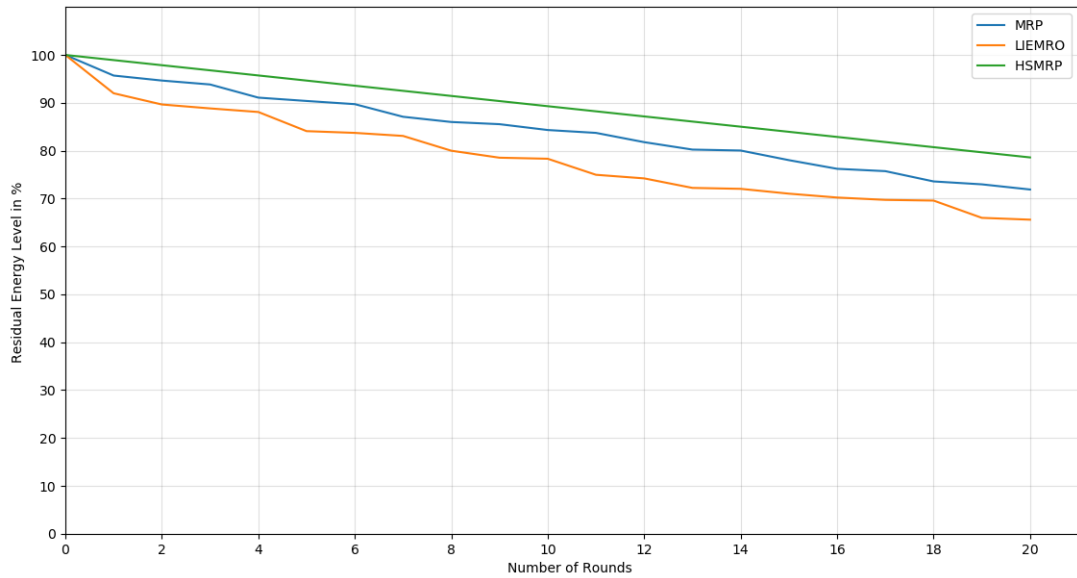


Figure 4.1: Average Residual Energy



| Residual Energy in Percentage |       |         |        |
|-------------------------------|-------|---------|--------|
| No. of rounds                 | HSMRP | MRP     | LIEMRO |
| 0                             | 100   | 100     | 100    |
| 1                             | 98.93 | 95.7    | 92     |
| 2                             | 97.86 | 94.65   | 89.66  |
| 3                             | 96.78 | 93.8375 | 88.83  |
| 4                             | 95.72 | 91.089  | 88.1   |
| 5                             | 94.65 | 90.39   | 84.09  |
| 6                             | 93.58 | 89.72   | 83.72  |
| 7                             | 92.51 | 87.9    | 83.09  |
| 8                             | 91.44 | 86      | 80     |
| 9                             | 90.37 | 85.54   | 78.54  |
| 10                            | 89.3  | 84.32   | 78.32  |
| 11                            | 88.23 | 83.72   | 74.97  |
| 12                            | 87.16 | 81.78   | 74.22  |
| 13                            | 86.09 | 80.24   | 72.24  |
| 14                            | 85.02 | 80.04   | 72.04  |
| 15                            | 83.95 | 78.03   | 71.03  |
| 16                            | 82.88 | 76.22   | 70.22  |
| 17                            | 81.81 | 75.73   | 69.73  |
| 18                            | 80.74 | 73.58   | 69.59  |
| 19                            | 79.67 | 72.98   | 65.98  |
| 20                            | 78.60 | 71.9    | 65.59  |

Table 4.2: Comparison Table on residual energy between MRP and HSMRP

## 4.2.2 Number of live nodes

It is the Number of live nodes that still remains after several transmission of data. As the sensor nodes are equipped with limited resource, nodes tends to die fast which will led to failure of the network. So, for every transmission the nodes that are frequently involve in the transmission of data will use up more energy compare to other nodes thus shortening the lifetime of the nodes which then leads to short lifetime of the network also. Here, we are comparing the results of the number of alive nodes remaining in the network after some rounds of data transmission . The nodes result is shown in the figure below . The figure shows the number of nodes in the proposed protocol tends to live longer as it uses only one primary path and one alternate path thus saving the energy of other nodes by putting them in sleep mode. The nodes around the sink/base station are the first to die as it is used up more frequently which result to some drop in the graph. The proposed protocol are seen to have more live nodes as compare to other protocol.

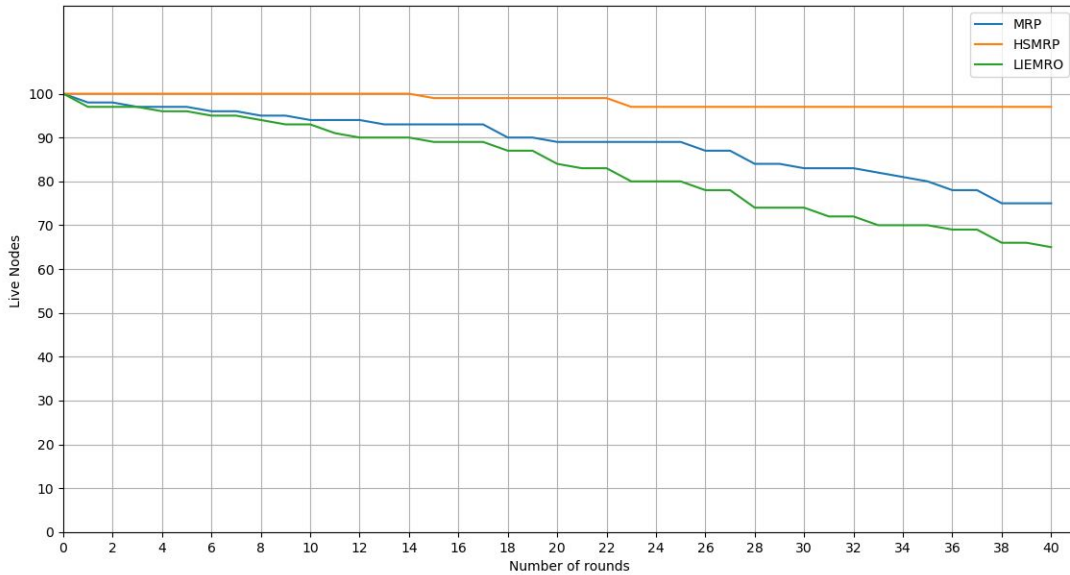


Figure 4.2: Number of live nodes

| Number of Live nodes |       |     |         |
|----------------------|-------|-----|---------|
| No. of rounds        | HSMRP | MRP | LIEMRO. |
| 0                    | 100   | 100 | 100     |
| 2                    | 100   | 98  | 97      |
| 4                    | 100   | 97  | 96      |
| 6                    | 100   | 96  | 95      |
| 8                    | 100   | 95  | 94      |
| 10                   | 100   | 94  | 93      |
| 12                   | 100   | 94  | 90      |
| 14                   | 100   | 93  | 89      |
| 16                   | 99    | 93  | 89      |
| 18                   | 99    | 90  | 87      |
| 20                   | 99    | 89  | 84      |
| 22                   | 99    | 89  | 83      |
| 24                   | 97    | 89  | 80      |
| 26                   | 97    | 87  | 78      |
| 28                   | 97    | 84  | 74      |
| 30                   | 97    | 83  | 72      |
| 32                   | 97    | 83  | 71      |
| 34                   | 97    | 81  | 70      |
| 36                   | 97    | 78  | 69      |
| 38                   | 97    | 76  | 66      |
| 40                   | 97    | 76  | 65      |

Table 4.3: Comparison table on number of live nodes between MRP and HSMRP

### 4.2.3 Packet Delivery Ratio

The packet delivery ratio is the ratio of packets successfully received to the total sent. Here, if the path fails, it discovers another path. As the protocols are all multipath based, there exist primary and alternate paths available for transmission of data when a path fails. Since the proposed protocol and MRP have efficient routing systems, the ratio for delivering is quite equivalent as compared to LIEMRO. Hence, the data loss is negligible in the MRP and in the proposed protocol (HSMRP). The result is shown in the figure below. The proposed protocol has a slight higher packet delivery ratio throughout the simulation.

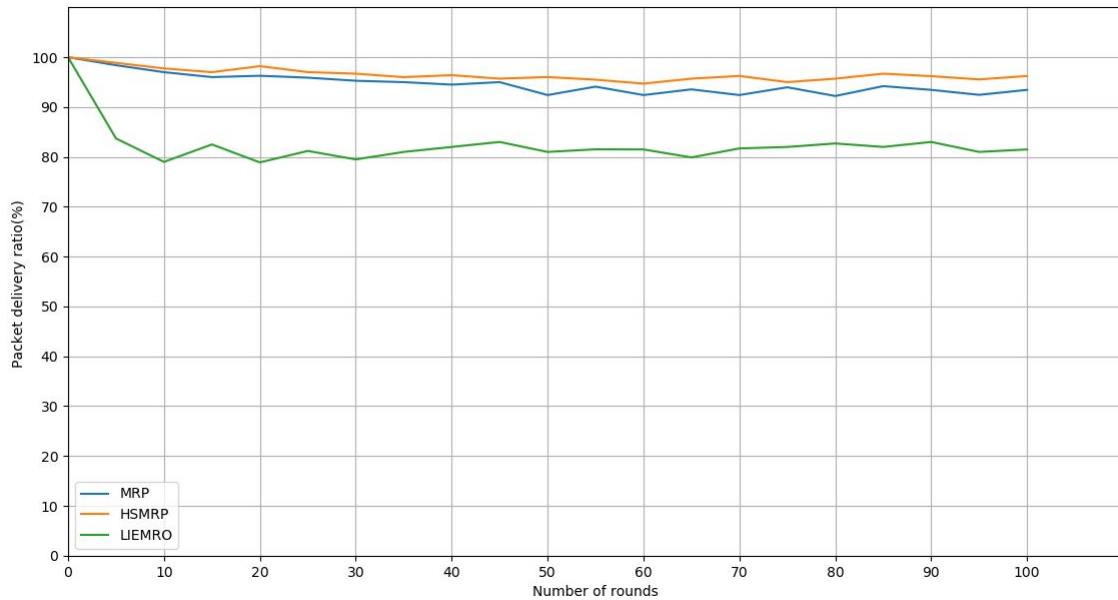


Figure 4.3: Packet Delivery Ratio

| Packet Delivery Ratio in Percentage |       |       |        |
|-------------------------------------|-------|-------|--------|
| No. of rounds                       | HSMRP | MRP   | LIEMRO |
| 0                                   | 100   | 100   | 100    |
| 5                                   | 98.87 | 98.4  | 83.7   |
| 10                                  | 97.75 | 97    | 79     |
| 15                                  | 97    | 96.01 | 82.5   |
| 20                                  | 98.21 | 96.28 | 78.9   |
| 25                                  | 97.01 | 95.9  | 81.2   |
| 30                                  | 96.7  | 95.27 | 79.5   |
| 35                                  | 96    | 95    | 81     |
| 40                                  | 96.4  | 94.5  | 82     |
| 45                                  | 95.7  | 95    | 83     |
| 50                                  | 96.02 | 92.41 | 81     |
| 55                                  | 95.5  | 94.1  | 81.52  |
| 60                                  | 94.7  | 92.41 | 81.5   |
| 65                                  | 95.7  | 93.55 | 79.9   |
| 70                                  | 96.24 | 92.41 | 81.7   |
| 75                                  | 95.01 | 93.97 | 82     |
| 80                                  | 95.7  | 92.22 | 82.7   |
| 85                                  | 96.7  | 94.2  | 82     |
| 90                                  | 96.2  | 93.45 | 83     |
| 95                                  | 95.55 | 92.45 | 81     |
| 100                                 | 96.23 | 93.45 | 81.5   |

Table 4.4: Comparison Table on packet delivery ratio between MRP and HSMRP

## 4.3 Advantages and Disadvantages of HSMRP

### **Advantage:**

In HSMRP, since the shortest path discovery is introduced the packet delivery ratio increases and as the number of nodes used to transmit the data is chosen via the shortest path, unnecessary nodes are not used up during the transmission of data. Thus, increasing the alive node of the network leading to a slow decrease in the residual energy of the nodes.

### **Disadvantage:**

Despite the advantages of HSMRP it also has disadvantages. Some disadvantages of HSMRP are that the nodes around the sink i.e. base station tend to die early due to continuous usage to complete data transmission as it is located near the sink. Nodes apart from the nodes near the sink, if participated in path discovery continuously might die early which fails to overcome in this protocol. It can be overcome by giving some constraint on how many times the node is being used.

# Chapter 5

## Conclusion

Wireless Sensor Networks are usually spread over large areas and are recently finding applications in many fields. With the current increasing usage of the Wireless Sensor Network application, various requirements also arises like requirement of methods which can manage the wireless sensor network's in a better way. The sensor nodes of the Wireless Sensor Networks are provided with limited capacity of batteries and resources. In Wireless Sensor Network the main challenges in the design of protocols is the energy efficiency of the network, as the sensor nodes are equipped with limited amount of energy. The main motive behind any routing protocol is to be as energy efficient as possible so as to keep the lifetime of the network running for a longer period of time.

In this paper we proposed hop count based shortest multipath routing protocol for WSN. This protocol is designed to increase the routing live node in the network, residual energy and decrease the consumption of energy through discovering multiple paths from the source to the destination. As it has sink initiated Route discovery process with the location information, the sink knows the location of the source. In this protocol, the path having the less hop count from the sink to the source is taken as primary path and the path having hop count nearest to the hop count of the primary

path is formed as alternate path. The primary path is used to transmit the data from source to the sink and if the path disrupts, then the alternate route is used for the purpose and if no path exists between the source and the sink the route discovery algorithm calls.

The simulation result finds the number of alive nodes, residual energy and average energy consumption. Among all these metrics we can see that the proposed performed better in all in comparison Which is the main motive of this paper.



# Chapter 6

## Future Scope

In Wireless sensor network, the main challenge is designing a protocol that are energy efficient as the power resource/amount of energy is limited in the nodes. The main motive behind any routing protocol is to be as energy efficient as possible so that the network lifetime runs for a longer period of time. In this paper we have presented a detailed description of HSMRP a protocol based on incorporating a threshold function to the existing protocol MRP. We have also presented a detailed comparison of MRP and HSMRP protocols. The new proposed protocol has future scope for betterment of increasing network lifetime.

But there is still many area for improvement. Like, the sensor nodes around the sink tends to die early since these nodes is always used to complete every data transmission and also nodes apart from the nodes near the sink, if used or participated in path discovery continuously it might die early too, improvement can be done in this part and many more. Due to time constraints on finishing this paper we were limited to implementing the mathematical formula on a single protocol. We urge further scholars in the field to undergo more research using different protocols and compare the improvements with the existing systems. Although many of these energy saving techniques look assuring , there are still many challenges that are still yet to be solved in the sensor networks.

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